

File Handling Interview Questions

Technical Interview Preparation — Question List

30 ESSENTIAL QUESTIONS

- Q1.** What is file handling in Python, and why is it important in real-world applications?
- Q2.** What is the difference between text files and binary files? When would you use each?
- Q3.** Explain the different file modes in Python: r, w, a, x, r+, w+, and a+.
- Q4.** What is the difference between w mode and a mode? How can choosing the wrong mode cause data loss?
- Q5.** What is the purpose of the with statement in file handling?
- Q6.** What happens to a file object when the with block is completed?
- Q7.** What is the difference between read(), readline(), and readlines()?
- Q8.** How does Python manage the file pointer while reading and writing a file?
- Q9.** What are seek() and tell()? Why are they useful in file handling?
- Q10.** Why is reading a very large file using read() sometimes inefficient? What is a better approach?
- Q11.** What is the difference between relative paths and absolute paths?
- Q12.** What is file encoding? Why is UTF-8 commonly used for text files?
- Q13.** What is the difference between flush() and close() in file handling?
- Q14.** What happens if a program terminates unexpectedly before buffered data is written to a file?
- Q15.** What are common exceptions during file handling? Explain FileNotFoundError, PermissionError, and IsADirectoryError.

Q16. What is serialization? Why is serialization required in programming?

Q17. What is deserialization? How is it different from normal file reading?

Q18. What is the difference between serialization of an object and simply writing its values into a text file?

Q19. What is the pickle module in Python, and why is it used?

Q20. What is the difference between `pickle.dump()` and `pickle.dumps()`?

Q21. What is the difference between `pickle.load()` and `pickle.loads()`?

Q22. What types of Python objects can generally be serialized using pickle?

Q23. What happens when you try to serialize an object that cannot be pickled?

Q24. What is the difference between JSON serialization and Pickle serialization?

Q25. Why is JSON generally preferred for data exchange between different programming languages?

Q26. Why can Pickle preserve Python-specific objects that JSON cannot directly represent?

Q27. Why is loading Pickle data from an untrusted source considered a security risk?

Q28. What is the difference between serialization, storage, and deserialization in a complete data-processing workflow?

Q29. What are the advantages and disadvantages of using serialization for storing application data?

Q30. In a real-world application, how would you decide between plain text, JSON, CSV, and Pickle for storing data?